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Gravity fed smoker

Abstract

A gravity fed smoker includes a smoking enclosure and an external stack. The stack is double walled with an inner wall, and an outer wall. A vented cooling space exists between the inner wall and the outer wall. An inner chamber of the external stack includes a fire box with a fire grate and a feed hopper positioned above the fire box. A smoke tunnel with a series of openings for releasing smoke into the smoking enclosure extends along the bottom of the food smoking enclosure and is connected to the external fire box at one end. Fuel, including charcoal, lump coal, or wood pellets, is loaded into the feed hopper. As the fuel burns on the fire grate and turns to ashes, the fuel is fed from the hopper onto the fire grate by gravity. A fan and dampers controls air flow through the fire grate and into the smoke tunnel.

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Background/Summary

CLAIM OF PRIORITY (1) This application claims priority from U.S. Provisional Patent Application Ser. No. 62/733,739, filed on Sep. 20, 2018, which is incorporated herein in its entirety.

FIELD OF THE INVENTION

(1) This invention relates to a smoker/grill for cooking food while imparting a smoked flavor to the food and more particularly to a smoker with an external gravity fed heat source.

BACKGROUND OF THE INVENTION

(2) Traditional smokers use a solid fuel, such as charcoal, lump coal, or wood pellets positioned inside the food smoking enclosure. Such an arrangement requires the user to open the lid or door of the smoker periodically during the smoking process in order to replenish the solid fuel and remove the ashes. Such replenishment of fuel and removal of ash represents a nuisance when the time to smoke food can extend for several hours. Periodically opening the door or lid of the smoking enclosure also allows the smoke to escape and disrupts the temperature within the smoking enclosure.

SUMMARY OF THE INVENTION

(3) A gravity fed smoker of the present invention eliminates the need to access the food smoking enclosure during the smoking process to replenish the solid fuel of the smoker or to remove ashes. Particularly, the gravity fed smoker includes a food smoking enclosure and an external stack. The food smoking enclosure is supported on four legs which in turn are supported on four wheels to facilitate movement of the smoker. The front of the food smoking enclosure includes a lid for gaining access to the interior of the food smoking closure. One or more grill racks are positioned within the food smoking enclosure for supporting the food to be cooked and smoked. A drip shield is supported on the bottom of the food smoking enclosure for catching drippings from the food being cooked and smoked and for funneling the drippings into a removable drip pan. The back of the food smoking enclosure includes an adjustable smoke outlet.

(4) The external stack is connected to one side of the food smoking enclosure. The external stack is generally rectangular in shape with four sides, a bottom, and a top opening. The stack is double walled with an inner wall and an outer wall. The inner wall defines an inner chamber. A cooling space exists between the inner wall and the outer wall. In a first embodiment, the outer wall is essentially continuous with circulation openings at the top and bottom. In a second embodiment the outer wall is slotted over most of its surface to assure that the surface of the outer wall remains at a safe temperature. The inner chamber of the external stack includes a fire box with a fire grate. The fire box is positioned near the bottom of the inner chamber of the external stack and adjacent an inlet opening of a smoke tunnel positioned along the bottom of the food smoking enclosure. The fire box has a smoke outlet that connects to the smoke inlet of the smoke tunnel.

(5) A fan with a fan outlet is connected to the fire box below the fire grate. In a first embodiment, the amount of air delivered by the fan to the fire grate can be controlled manually or automatically either by use of a variable speed fan or by varying the duty cycle of a constant speed fan. In a second embodiment, a check flapper is positioned in the fan outlet to ensure that hot air does not flow from the fire box to the fan when the fan is shut off. In addition, a manual combustion air

damper is positioned in the fan outlet to control the amount of air from the fan to the fire box and to shut off air to the fire box after cooking has been completed and the fan is turned off. Shutting off air to the fire box after cooking has been completed serves to extinguish the fuel in the fire box and thereby conserve fuel.

(6) As previously stated, the smoke tunnel extends along the bottom of the food smoking enclosure from side to side. The smoke tunnel inlet at one end of the smoke tunnel extends through one side of the food smoking enclosure. The other end of the smoke tunnel is closed. A series of openings in the sides of the smoke tunnel extend along the length of the smoke tunnel to allow smoke and heat to flow from the smoke inlet of the smoke tunnel into the food smoking enclosure and out the smoke outlet of the food smoking enclosure. In a first embodiment of the present invention, the openings in the smoke tunnel are progressively larger starting from the smoke inlet opening of the smoke tunnel in order to evenly distribute the smoke and heat into the food smoking enclosure. In a second embodiment of the present invention the openings in the smoke tunnel has the larger openings at the inlet and the closed end of the smoke tunnel with smaller openings near the middle of the smoke tunnel. The second embodiment also includes a smoke damper that controls the amount of smoke flowing from the fire box into the smoke tunnel. Closing the smoke damper after cooking has been completed serves to extinguish the fuel in the fire box and thereby conserve fuel.

(7) An ash pan is positioned in the inner chamber of the stack below the fire grate to catch the ashes from the solid fuel burning on the fire grate. The ash pan is also designed to receive larger wood chunks that will smolder under the falling embers. This allows additional flavoring of the smoke that permeates the food smoking enclosure. The stack has an access door near the bottom of the stack adjacent the ash pan for removal of the ash pan as necessary and during startup, access for lighting the fuel supported on the fire grate.

(8) The inner chamber also includes a feed hopper for containing solid fuel, such as charcoal, lump coal, or wood pellets. The feed hopper is positioned above and connected to the top of the fire box. The inner chamber of the stack has a top opening for loading solid fuel into the feed hopper. The stack has a top lid that closes the top opening and seals the inner chamber. In the first embodiment, the lid has an adjustable lid vent to control the burn rate of the solid fuel. Also, in the first embodiment, the outer wall of the external stack has top vents near the top of the stack and bottom vents in the bottom of the stack that communicate with the cooling space so that air can circulate from the bottom vents, through the cooling space, and out the top vents to cool the inner wall of the stack and keep the outer stack wall at a safe temperature. In the second embodiment, the outer stack wall of the external stack has multiple slots in order to provide maximum air flow to the inner stack wall and thereby to keep the outer stack wall at a safe temperature.

(9) A utility tray is attached on one external side of the food smoking enclosure to provide a place for holding food and utensils. In the first embodiment, a warming tray is attached to one side of the stack and utilizes the heat from the stack to keep food warm that has already been cooked or to warm plates in anticipation of serving the food.

(10) In operation, the feed hopper is loaded with solid fuel in the form of charcoal, lump coal, or wood pellets. The solid fuel descends by gravity and rests on the fire grate of the fire box. By means of the access door at the bottom of the stack, the user can access the bottom of the fire grate and light the solid fuel on the fire grate. During the lighting process and initial startup, the lid vent in the first embodiment may be opened and the fan may be run at a high speed in order to rapidly increase combustion and initial startup temperature. Once the fuel has been lit, and the fire box has reached an initial high startup temperature, the lid vent is closed, and the fan speed is reduced. Alternatively, in the second embodiment, the fan can operate at a fixed speed, and the amount of air supplied to the fire box is controlled by the combustion air damper in the fan outlet.

(11) During the cooking and smoking operation, the fan speed, in the first embodiment, is controlled to provide air below the fire grate to maintain combustion and to force the resulting smoke and heat out of the fire box smoke outlet, into the smoke tunnel inlet, out of the smoke

tunnel openings, and into the food smoking enclosure. In the second embodiment, the amount of air delivered to the fire box is controlled by the combustion air damper in the fan outlet. Because after initial startup the inner chamber of the stack is sealed by the lid and the closed lid vent, the solid fuel does not burn up the stack. Instead combustion is controlled by the volume of air delivered by the fan as the solid fuel burns just above the fire grate, and the ash drops through the fire grate into the ash pan below. As the ash drops through the fire grate, the solid fuel above the fire grate drops under the force of gravity onto the fire grate to replace the spent fuel and ash. The falling ash and embers will cause additional wood chunks in the ash pan to smolder providing for additional flavor profiles.

(12) In the first embodiment with the lid vent open and the fan running at a fairly high-speed and air volume, the grill rack inside the food smoking enclosure initially can be raised to a temperature sufficient to sear meat placed on the grill rack (approximately 1000° F.). Subsequently, by closing the lid vent and adjusting the air flow from the fan downward, the temperature inside the food smoking enclosure can be lowered to slowly and continuously cook and smoke delicate foods (cheese and fish) in the food smoking enclosure (140° F.). The temperature can also be controlled by opening the lid of the food smoking enclosure. In the second embodiment, the internal temperature in the food smoking enclosure is initially raised to a high temperature by running the fan at fairly high-speed with the combustion air damper open and the smoke damper open. Subsequently, the temperature in the food smoking enclosure can be modulated by adjusting the combustion air damper and or the smoke damper as well as the fan speed.

(13) Further objects, features and advantages will become apparent upon consideration of the following detailed description of the invention when taken in conjunction with the drawings and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front perspective view of a first embodiment of a gravity fed smoker in accordance with the present invention.
- (2) FIG. 2 is a front elevation view of the first embodiment of the gravity fed smoker in accordance with the present invention.
- (3) FIG. 3 is a back elevation view of the first embodiment of the gravity fed smoker in accordance with the present invention.
- (4) FIG. 4 is a top plan view of the first embodiment of the gravity fed smoker in accordance with the present invention.
- (5) FIG. 5 is a right side elevation view of the first embodiment of the gravity fed smoker in accordance with the present invention.
- (6) FIG. 6 is a bottom plan view of the first embodiment of the gravity fed smoker in accordance with the present invention.
- (7) FIG. 7 is a front elevation section view of the first embodiment of the gravity fed smoker as seen along line 7-7 of FIG. 5 in accordance with the present invention.
- (8) FIG. 8 is a front perspective section view of the first embodiment of the gravity fed smoker as seen along line 8-8 of FIG. 5 in accordance with the present invention.
- (9) FIG. 9 is a front perspective view of a second embodiment of a gravity fed smoker in accordance with the present invention.
- (10) FIG. 10 is a right side perspective view of the second embodiment of the gravity fed smoker in accordance with the present invention.
- (11) FIG. 11 is a front elevation view of the second embodiment of the gravity fed smoker in accordance with the present invention.

- (12) FIG. 12 is a right side elevation view of the second embodiment of the gravity fed smoker in accordance with the present invention.
- (13) FIG. 13 is a back elevation view of the second embodiment of the gravity fed smoker in accordance with the present invention.
- (14) FIG. 14 is a top plan view of the second embodiment of the gravity fed smoker in accordance with the present invention.
- (15) FIG. 15 is a left side elevation view of the second embodiment of the gravity fed smoker in accordance with the present invention.
- (16) FIG. 16 is a bottom plan view of the second embodiment of the gravity fed smoker with the cart tray removed to reveal detail in accordance with the present invention.
- (17) FIG. 17 is a front elevation view of a second embodiment of the gravity fed smoker with portions of the food smoking enclosure removed to reveal internal detail in accordance with the present invention.
- (18) FIG. 18 is a front perspective section view of the second embodiment of the gravity fed smoker with portions of the fire box and hopper removed to show internal detail as seen along line 18-18 of FIG. 12 in accordance with the present invention.
- (19) FIG. 19 is a front elevation view of the second embodiment of the gravity fed smoker with portions of the food smoking enclosure, the fire box, and the hopper removed to show internal detail in accordance with the present invention.
- (20) FIG. 20 is a right side perspective view of the second embodiment of the gravity fed smoker with the door and portions of the fire box and the hopper removed to show internal detail in accordance with the present invention.
- (21) FIG. 21 is an enlarged right side perspective view of the second embodiment of the gravity fed smoker with the door and portions of the fire box and hopper removed to show internal detail in accordance with the present invention.
- (22) FIG. 22 is an enlarged front perspective view of the second embodiment of the gravity fed smoker with the door and portions the fire box and hopper removed to show internal detail in accordance with the present invention.
- (23) FIG. 23 is an enlarged front perspective view of the second embodiment of the gravity fed smoker with the door and portions the fire box and hopper removed to show internal detail in accordance with the present invention.
- (24) FIG. 24 is an enlarged front perspective view of the second embodiment of the gravity fed smoker with the door and portions the fire box and hopper removed to show internal detail in accordance with the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(25) FIGS. 1-8 illustrate a first embodiment of the present invention. FIGS. 9-24 illustrate a second embodiment of the present invention. Turning to FIG. 1, a gravity fed smoker 10 includes a food smoking enclosure 12 and an external stack 42. The food smoking enclosure 12 has a front 32, a back 20, a left side 18, a right side 16, a top 14, and a bottom 22. The food smoking enclosure 12 is supported on four legs 34 which in turn are supported on four wheels 36 to facilitate movement of the gravity fed smoker 10. A cart tray 37 extends between the legs below the foods smoking enclosure to provide stability and extra storage space. The front 32 includes a lid 28 for gaining access to the interior of the food smoking enclosure 12. A grill rack 40 (FIG. 7) is supported on the grill rack supports 41 (FIG. 8) within the food smoking enclosure 12. The grill rack 40 in turn supports the food to be cooked and smoked. In FIG. 8 the grill rack 40 has been removed for the sake of clarity. A drip pan (not shown) is supported on the bottom 22 of the food smoking enclosure 12 for catching drippings from the food being cooked and smoked. The back 20 of the food smoking enclosure includes an adjustable smoke outlet 24.

(26) A smoke tunnel 86 extends along the bottom 22 of the food smoking enclosure 12 from the right side 16 to the left side 18. One end of the smoke tunnel 86 has a smoke inlet 90 (FIG. 8)

through the right side **16** of the food smoking enclosure **12**. The other end **87** of the smoke tunnel **86** is closed. A series of openings **88** in the sides of the smoke tunnel **86** extend along the length of the smoke tunnel **86** to allow smoke and heat to flow from the smoke inlet **90** of the smoke tunnel **86** into the food smoking enclosure **12** and out the smoke outlet **24** of the food smoking enclosure **12**. The openings **88** in the smoke tunnel **86** are progressively larger starting from the smoke inlet **90** of the smoke tunnel **86** in order to evenly distribute the smoke and heat from the smoke tunnel **86** into the food smoking enclosure **12**. By varying the size and spacing of the openings **88** and by varying the dimensions of the smoke tunnel **86**, the smoke tunnel **86** can be tuned to distribute smoke and heat evenly into the smoking enclosure **12** to assure consistent cooking and smoking of the food within the smoking enclosure **12**.

(27) The external stack **42** is connected to the right side **16** of the food smoking enclosure **12**. The external stack **42** is generally rectangular in shape with a front **46**, a left side **48**, a right side **50**, a back **52**, a bottom **54** (FIG. 6), and a top opening **56**. The stack **42** includes an upper feed hopper **44**, a fire box **76**, and a lower stack portion **83**. The stack **42** is double walled with an inner wall **60** and an outer wall **62**. The inner wall **60** defines an inner chamber **64**. A cooling space **66** exists between the inner wall **60** and the outer wall **62**. The inner chamber **64** of the external stack **42** includes the fire box **76** with a bottom fire grate **78**. The fire box **76** is positioned toward the bottom **54** of the inner chamber **64** of the external stack **42** and adjacent to the smoke tunnel inlet **90** of the smoke tunnel **86**. The fire box **76** has a smoke outlet **82** covered by a smoke outlet grate **89** (FIG. 8) that connects to the smoke tunnel inlet **90** of the smoke tunnel **86**.

(28) A fan **92** with a fan outlet **94** (FIG. 8) is connected to the inner chamber **64** below the fire grate **78**. The amount of air delivered by the fan **92** to the fire grate **78** can be controlled manually or automatically either by use of a variable speed fan or by varying the duty cycle of a constant speed fan.

(29) An ash pan **80** is located in the lower portion **83** of the stack **42** and is supported on the bottom **54** of the inner chamber **64** of the stack **42** below the fire grate **78**. The ash pan **80** receives the ash and embers from the fire grate **78**. The ash pan **80** also can be loaded with additional wood chunks that will smolder during the cooking and smoking process to add additional smoke flavors. The stack **42** has a lower access door **84** near the bottom **54** of the stack **42** adjacent the ash pan **80** for removal of the ash pan **80** as necessary and for gaining access below the fire grate **78** in order to light the solid fuel supported on the fire grate **78**.

(30) The feed hopper **44** is located near the top of the stack **42** and contains the solid fuel, such as charcoal, lump coal, or wood pellets. The feed hopper **44** is positioned above and connected to the top of the fire box **76**. The inner chamber **64** of the stack **42** has a top opening **56** for loading fuel into the feed hopper **44**. The stack **42** has a top lid **58** that closes the top opening **56** and seals the inner chamber **64**. The top lid **58** also includes a lid vent **72** that can be opened and closed in connection with the startup of combustion at the fire grate **78**. The outer stack wall **62** has top vents **68** near the top of the stack **42** and bottom vents **70** (FIG. 6) in the bottom **54** of the stack **42** that communicate with the cooling space **66** between the inner stack wall **60** and the outer stack wall **62** so that air can circulate from the bottom vents **70**, through the cooling space **66**, and out the top vents **68** to cool the inner wall **60** of the stack **46** and keep their outer stack wall **62** at a safe temperature.

(31) A utility tray **38** is attached on the left side **18** of the food smoking enclosure **12** to provide a place for holding food and utensils. A warming tray **74** is attached to the right side **50** of the stack **42** and utilizes the heat from the stack **42** to keep food warm that has already been cooked or to warm plates in anticipation of serving the food.

(32) In operation, the feed hopper **44** is loaded with solid fuel in the form of charcoal, lump coal, or wood pellets. The solid fuel in the feed hopper **44** descends by gravity and comes to rest on the fire grate **78** of the fire box **76**. By means of the lower access door **84** at the bottom **54** of the stack **42**, the user can access the bottom of the fire grate **78** and light the solid fuel on the fire grate **78**. Once

the solid fuel has been lit and in order to increase the temperature inside the food smoking enclosure **12** rapidly at startup, the lid vent **72** can be opened, and the fan **92** run at a relatively high speed in order to accelerate combustion within the fire box **76**. Once the temperature inside the food smoking enclosure **12** has reached the initial desired high temperature for searing meat in the grill rack **40**, the lid vent **72** is closed thereby sealing the inner chamber **64**, and the fan **92** speed is controlled to provide air below the fire grate **78** to maintain combustion. The fan **92** forces the resulting smoke and heat out of the fire box smoke outlet **82**, through the smoke tunnel inlet **90**, into the smoke tunnel **86**, out of the smoke tunnel openings **88**, and into the food smoking enclosure **12**. Because the inner chamber **64** of the stack **42** is sealed by the stack lid **58** and the closed lid vent **72**, the solid fuel does not burn up the inner chamber **64**. Instead as the solid fuel burns just above the fire grate **78**, the ash drops through the fire grate **78** into the ash pan **80** below. As the ash drops through the fire grate **78**, the fuel above the fire grate **78** drops under the force of gravity onto the fire grate **78** to replace the spent fuel and ash.

(33) Initially with the rapid startup described above, the grill rack **40** inside the food smoking enclosure **12** can be raised to a temperature between 700° F. and 1000° F. in order to sear meat placed on the grill rack **40**. Subsequently, by adjusting the air flow from the fan **92** downward, the temperature inside the food smoking enclosure **12** can be lowered to as low as 140° F. to slowly and continuously cook and smoke the food in the food smoking enclosure **12**. The temperature can also be controlled by opening the lid **28** of the food smoking enclosure **12**.

(34) The smoker **10** includes a temperature probe and a controller (not shown) that controls the duty cycle or the speed rate of the fan **92**. The desired temperature set point is entered into the controller, and the controller drives the speed of the fan **92** to reach the set temperature point. Once the temperature set point has been reached, the controller will reduce the fan's speed to maintain the set point temperature. By stopping the fan, the combustion at the fire grate **78** will shut down, and the solid fuel left in the hopper **44** can be saved the next cooking and smoking operation.

(35) A second embodiment of the present invention is shown in FIGS. 9-24. The second embodiment is a gravity fed smoker **110** that includes a food smoking enclosure **112** and an external stack **142**. The food smoking enclosure **112** has a front **132**, a back **120**, a left side **118**, a right side **116**, a top **114**, and a bottom **122**. The food smoking enclosure **112** is supported on four legs **134** which in turn are supported on four wheels **136** to facilitate movement of the gravity fed smoker **110**. The front **132** includes a lid **128** for gaining access to the interior of the food smoking enclosure **112**. A grill rack **140** (FIGS. 17-20) is supported on the grill rack supports **141** (FIG. 18) within the food smoking enclosure **112**. The grill rack **140** in turn supports the food to be cooked and smoked. A drip shield **125** is supported on the bottom **122** of the food smoking enclosure **112** for catching drippings from the food being cooked and smoked. The drip shield **125** funnels the drippings into a removable drip pan **126**. The back **120** of the food smoking enclosure includes an adjustable smoke outlet **124**.

(36) A smoke tunnel **186** extends along the bottom **122** of the food smoking enclosure **112** from the right side **116** to the left side **118**. One end of the smoke tunnel **186** has a smoke inlet **190** (FIG. 22) through the right side **116** of the food smoking enclosure **112**. The other end **187** of the smoke tunnel **186** is closed. A series of openings **188** in the sides of the smoke tunnel **186** extend along the length of the smoke tunnel **186** to allow smoke and heat to flow from the smoke inlet **190** of the smoke tunnel **186** into the food smoking enclosure **112** and out the smoke outlet **124** of the food smoking enclosure **112**. The openings **188** in the smoke tunnel **186** are larger at each end and smaller in the middle the smoke tunnel **186**. The smoke tunnel **186** is topped by a triangular-shaped cap **185**. By varying the size and spacing of the openings **188** and by varying the dimensions of the smoke tunnel **186**, the smoke tunnel **186** can be tuned to distribute smoke and heat evenly into the smoking enclosure **112** to assure consistent cooking and smoking of the food within the smoking enclosure **112**.

(37) The external stack **142** is connected to the right side **116** of the food smoking enclosure **112**.

The external stack **142** is generally rectangular in shape with a front **146**, a left side **148**, a right side **150**, a back **152**, a bottom **154**, and a top opening **156** (FIG. **18**). The stack **142** includes an upper feed hopper **144**, a fire box **176**, and a lower stack portion **183**. The stack **142** is double walled with an inner wall **160** and an outer wall **162**. The inner wall **116** defines an inner chamber **164** (FIG. **18**). A cooling space **166** exists between the inner wall **160** and the outer wall **162**. In order to ensure a safe temperature on the outer wall **162**, the outer wall **162** consists of a series of slots allowing free movement of air in the cooling space **166**. Particularly, the slots in the outer wall **162** result in the outer wall **162** having an open percentage of 40% to 75%, depending on temperatures of hopper and desired skin temperature.

(38) The fire box **176** is positioned below the feed hopper **144** and above the lower stack portion within the inner chamber **164** of the external stack **142**. A fire grate **178** is positioned at the bottom of the fire box **176**. The fire box **176** is lined with fire brick **177** to protect the inner wall **160** and lessen the amount of heat transferred into the cooling space **166**. The fire box **176** has a smoke outlet **182** that connects to the smoke tunnel inlet **190** of the smoke tunnel **186**. A slidable manual smoke damper **195** is interposed between the smoke outlet **182** of the fire box **176** and the smoke tunnel inlet **190** of the smoke tunnel **186**. The smoke damper **195** slides between an open position in which smoke can flow from the fire box **176** into the smoke tunnel **186** and a closed position in which smoke cannot flow from the fire box **176** into the smoke tunnel **186**. The smoke damper **195** can be manually adjusted to any intermediate position between the open position and the closed position.

(39) An angled grate **189**, which forms part of the fire box outlet **182**, is positioned on the fire box side of the smoke damper **195** as shown in FIG. **21**. The angled grate **189** catches any embers or other solid pieces of fuel that might be created in the fire box and propelled toward the smoke outlet **182** and the smoke damper **195** by air flow created by and a fan **192** (FIG. **22**). Once an ember is caught by the angled grate **189**, the ember slides down the angled grate **189** and back into the fire box **176**. Consequently, angled grate **189** keeps live embers from passing through the smoke outlet **182** and into the smoke tunnel inlet **190** and the smoke tunnel **186**.

(40) The fan **192** is mounted to a fan tunnel **191** which in turn is mounted to the stack **142**. The fan tunnel **191** has a fan outlet **194** that is connected to the inner chamber **164** below the fire grate **178**. A check flapper **193** is pivotally mounted in the fan tunnel **191**. The check flapper **193** allows air to flow from the fan **192** into the fire box **176** but precludes air from flowing from the fire box **176** toward the fan **192**. A slidable manual combustion air damper **197** is positioned at the fan outlet **194**. The combustion air damper **197** slides between an open position in which combustion air can flow from the fan **192** into the fire box **176** and a closed position in which combustion air cannot flow from the fan **192** into the fire box **176**. The combustion air damper **197** can be manually adjusted to any intermediate position between the open position and the closed position. The amount of air delivered by the fan **192** to the fire box **176** can be controlled by varying the speed of the fan or by manually adjusting the combustion air damper **197** between the open position, the closed position, and any intermediate position.

(41) An ash pan **180** is supported on the bottom **154** of the inner chamber **164** of the stack **142** below the fire grate **178**. The ash pan **180** receives the ash and embers from the fire grate **178**. The ash pan **180** also can be loaded with additional wood chunks that will smolder during the cooking and smoking process to add additional smoke flavor. The stack **142** has a lower access door **184** near the bottom **154** of the stack **142** adjacent the ash pan **180** for removal of the ash pan **180** as necessary and for lighting the solid fuel supported on the fire grate **178**.

(42) The inner chamber **164** also includes a feed hopper **144** for containing the solid fuel, such as charcoal, lump coal, or wood pellets. The feed hopper **144** is positioned above and connected to the top of the fire box **176**. The inner chamber **164** of the stack **142** has a top opening **156** (FIG. **18**) for loading fuel into the feed hopper **144**. The stack **142** has a top lid **158** that closes the top opening **156** and seals the inner chamber **164**.

- (43) A utility tray **138** is attached on the left side **118** of the food smoking enclosure **112** to provide a place for holding food and utensils. An additional storage shelf **137** is provided below the food smoking enclosure **112**.
- (44) In operation, the feed hopper **144** is loaded with solid fuel in the form of charcoal, lump coal, or wood pellets. The solid fuel in the feed hopper **144** descends by gravity and comes to rest on the fire grate **178** of the fire box **176**. By means of the lower access door **184** at the bottom of the stack **142**, the user can access the bottom of the fire grate **178** and light the solid fuel on the fire grate **178**. Once the solid fuel has been lit and in order to increase the temperature inside the food smoking enclosure **112** rapidly at startup, the combustion air damper **197** is opened, the fan **192** run at a relatively high speed in order to accelerate combustion within the fire box **176**, and the smoke damper **195** is open to facilitate air circulation through the fire grate **178**. Once the temperature inside the food smoking enclosure **112** has reached the initial desired high temperature for searing meat in the grill rack **140**, the cooking temperature inside the food smoking enclosure **112** can be controlled by adjusting the fan speed, manually adjusting the combustion air damper **197** and manually adjusting the smoke damper **195**. Because the inner chamber **164** of the stack **142** is sealed by the stack lid **158** and by limiting the combustion air, the solid fuel does not burn up the inner chamber **164**. Instead as the solid fuel burns just above the fire grate **178**, the ash drops through the fire grate **178** into the ash pan **180** below. As the ash drops through the fire grate **178**, the fuel above the fire grate **178** drops under the force of gravity onto the fire grate **178** to replace the spent fuel and ash.
- (45) The smoker **110** includes an enclosure temperature sensor **185**, a temperature probe **199**, and a controller **139** that controls the speed of the fan **192**. The desired temperature set point for the smoking enclosure **112** is entered into the controller, and the controller will drive the fan **192** to reach the set temperature point. Once enclosure temperature sensor **185** reaches the temperature set point, the controller **139** will reduce the fan's speed to maintain the set point temperature. The controller **139** also determines when the cooking process has been completed based on receiving temperature data from the temperature probe **199**.
- (46) Once the cooking process has been completed, the fan **192** is turned off, and the smoke damper **195** and the combustion air damper **197** are closed. With the fan **192** turned off, the smoke damper **195** closed, and the combustion air damper **197** closed, the inner chamber **164** of the stack **142** is substantially deprived of combustion air. Without combustion air, the fire in the fire box **176** is extinguished thereby saving fuel in the feed hopper **144** and the top of the fire box **176**.
- (47) While this invention has been described with reference to preferred embodiments thereof, it is to be understood that variations and modifications can be affected within the spirit and scope of the invention as described herein and as described in the appended claims.

Claims

1. An apparatus for cooking and smoking a food product comprising: a. an enclosure including: i. at least one cooking rack located within the enclosure for holding the food product to be cooked, the at least one cooking rack positioned above a bottom area within the enclosure; and ii. a smoke inlet for introducing smoke and heat into the enclosure; b. a stack external to the enclosure, the stack comprising an opening and the stack further comprising: i. a fire box for combusting fuel, the fire box located below the opening including: (a) a fire box outlet configured to permit smoke and heat to be introduced from the combusting fuel into the enclosure through the smoke inlet; ii. a fuel hopper positioned for gravity feeding fuel into the fire box; c. a smoke tunnel positioned within the bottom area of the enclosure, the smoke tunnel comprising a hollow structure having a first distal end and a second distal end located opposite each other along a horizontal axis within the bottom area of the enclosure, the hollow structure of the smoke tunnel further comprising at least one outer wall that surrounds at least a portion of a hollow space extending along the horizontal direction

- within the bottom area of the enclosure, wherein one or more openings are positioned along the outer wall; and d. a fan configured to deliver air from outside of the apparatus into the fire box to combust the fuel and create a flow of air from the combusting fuel in the fire box, through the fire box outlet, through the smoke inlet and into the hollow space of the smoke tunnel, and out of the one or more openings of the smoke tunnel to the cooking grate.
2. The apparatus of claim 1, wherein the stack includes an inner stack wall and an outer stack wall with a cooling space between the inner stack wall and the outer stack wall.
 3. The apparatus of claim 2, wherein the outer stack wall has a series of openings configured to allow free movement of air within the cooling space.
 4. The apparatus of claim 3, wherein openings in the outer wall result in the outer wall having an open percentage of from 40% to 75%.
 5. The apparatus of claim 1, wherein the fire box outlet includes an adjustable smoke damper for controlling the amount of smoke generated in the fire box that passes through the fire box outlet.
 6. The apparatus of claim 1, wherein the fire box outlet includes an angled grate for catching embers or pieces of fuel and precluding such embers or pieces of fuel from entering the smoke inlet.
 7. The apparatus of claim 1, further comprising a fan tunnel with a fan outlet configured to provide air from the fan into the fire box and wherein the fan tunnel further comprises a check flapper.
 8. The apparatus of claim 7, wherein the fan outlet includes an air damper configured to control an amount of air delivered to the fire box from the fan.
 9. The apparatus of claim 1, wherein the amount of air delivered by the fan to the fire box is controlled at least in part by adjusting fan speed.
 10. The apparatus of claim 1, wherein the amount of air delivered by the fan to the fire box is controlled at least in part by cycling the fan on and off.
 11. An apparatus for cooking and smoking a food product comprising: a. an enclosure including a smoke inlet; and b. a stack external to the enclosure, the stack comprising: i. a fire box for combusting fuel, the fire box comprising a fire grate and a fire box outlet, the fire box configured to provide smoke and heat from the combusting fuel through the fire box outlet to the smoke inlet of the enclosure; ii. a fuel hopper comprising an opening for receiving fuel, the fuel hopper configured to gravity feed the fuel into the fire box; and c. a smoke tunnel positioned within a bottom area of the enclosure, the smoke tunnel comprising a hollow structure having a first distal end and a second distal end located opposite each other along a horizontal axis within the bottom area of the enclosure, such that when the smoke tunnel is positioned within the bottom area of the enclosure, the smoke tunnel extends in a horizontal direction across the bottom area of the enclosure, the hollow structure of the smoke tunnel further comprising at least one outer wall that surrounds at least a portion of a hollow space extending along the horizontal direction within the bottom area of the enclosure, wherein one or more openings are positioned along the outer wall; and d. a fan configured to deliver air from outside of the apparatus into the fire box to combust the fuel and create a flow of air from the combusting fuel in the fire box, through the fire box outlet, through the smoke inlet and into the hollow space of the smoke tunnel, and out of the one or more openings of the smoke tunnel within the bottom area of the enclosure.
 12. The apparatus of claim 11, wherein the fan is disposed at least partially within a fan tunnel, the fan tunnel mounted to the stack and comprising a fan outlet.
 13. The apparatus of claim 12, further comprising a check flapper disposed within the fan tunnel.
 14. The apparatus of claim 11, further comprising a controller operatively coupled to the fan and configured to control the temperature within the enclosure.
 15. The apparatus of claim 11, where the fire box further comprises a fire grate disposed in a bottom portion of the fire box.
 16. The apparatus of claim 15, further comprising an ash pan disposed below the fire grate.
 17. The apparatus of claim 15, wherein the fire box further comprises an angled grate disposed

between the fire box outlet and the fire box.

18. An apparatus comprising: a. an enclosure for cooking and smoking food, the enclosure comprising a cooking rack and a bottom area below the cooking rack; b. a stack external to the enclosure, the stack comprising: i. a fuel hopper comprising an opening for receiving fuel; ii. a fire box configured to receive fuel from the fuel hopper, wherein the fire box is further configured to provide smoke and heat to the enclosure; and c. a smoke tunnel positioned within the bottom area of the enclosure, the smoke tunnel comprising: i. a first end and a second end separated by a length along the smoke tunnel; ii. at least one side wall that defines a hollow space for receiving smoke and heat from the fire box between the first and second ends; and iii. a plurality of openings within the at least one side wall for delivering smoke and heat into the bottom area of the enclosure, wherein the plurality of openings are positioned on the at least one side wall along at least half of the length between the first end and the second end of the smoke tunnel; d. a fan configured to deliver air from outside of the apparatus into the fire box to combust the fuel and create a flow of air from the combusting fuel in the fire box, into the hollow space of the smoke tunnel, and out of the plurality of openings of the smoke tunnel within the bottom area of the enclosure.

19. The apparatus of claim 18, wherein at least one of the plurality of openings positioned along the at least one side wall of the smoke tunnel is a different size than at least another opening of the plurality of openings.

20. The apparatus of claim 18, wherein the smoke tunnel further comprises a top wall that is triangular-shaped.

21. The apparatus of claim 20, wherein the hollow space has a pentagonal-shaped cross-section.

22. The apparatus of claim 18, wherein the first end and second end of the smoke tunnel are separated by the length along a horizontal axis.
